

Testing for Externalities in Network Formation by Simulation

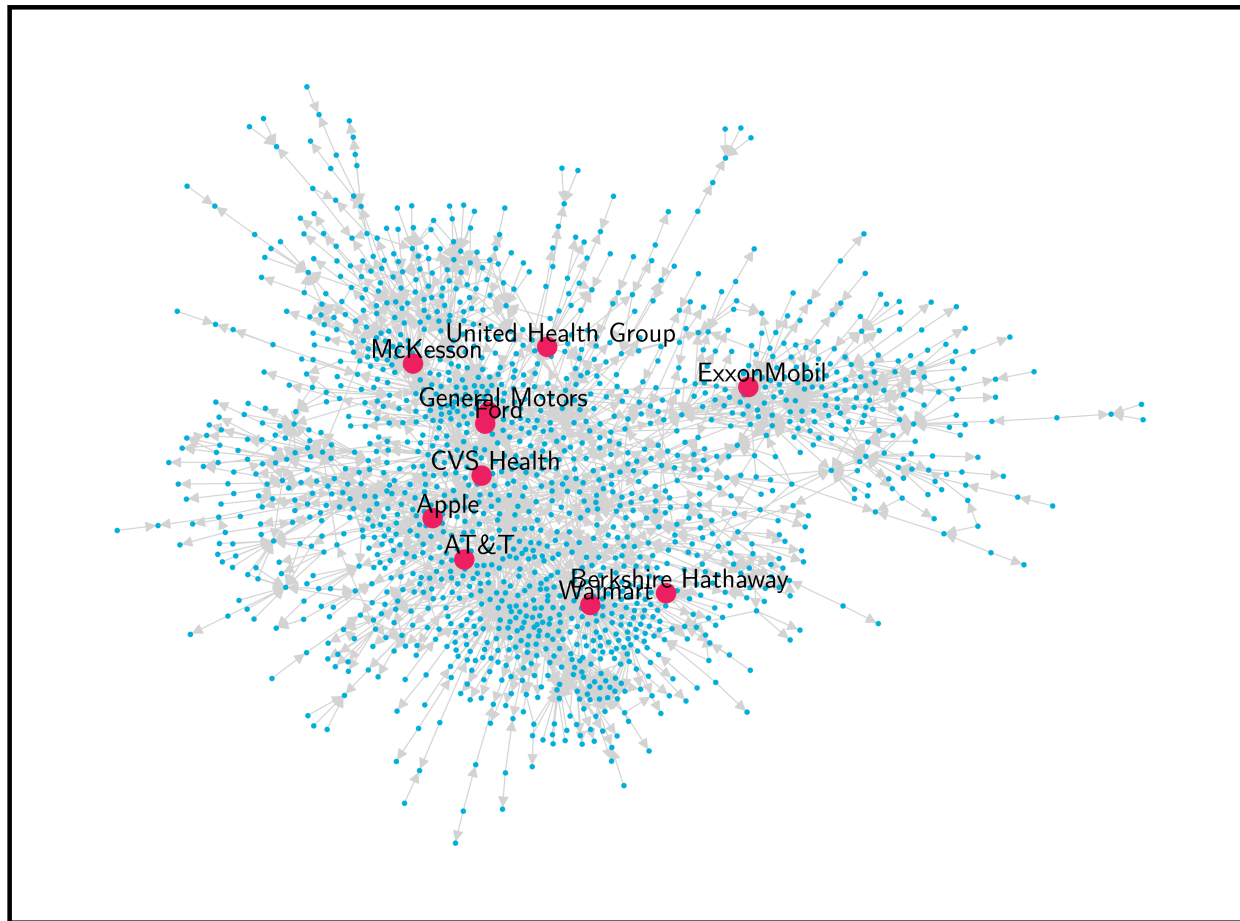
Econometric Methods for Social Spillovers and Networks

Universität St.Gallen, September 21st to 25th, 2026

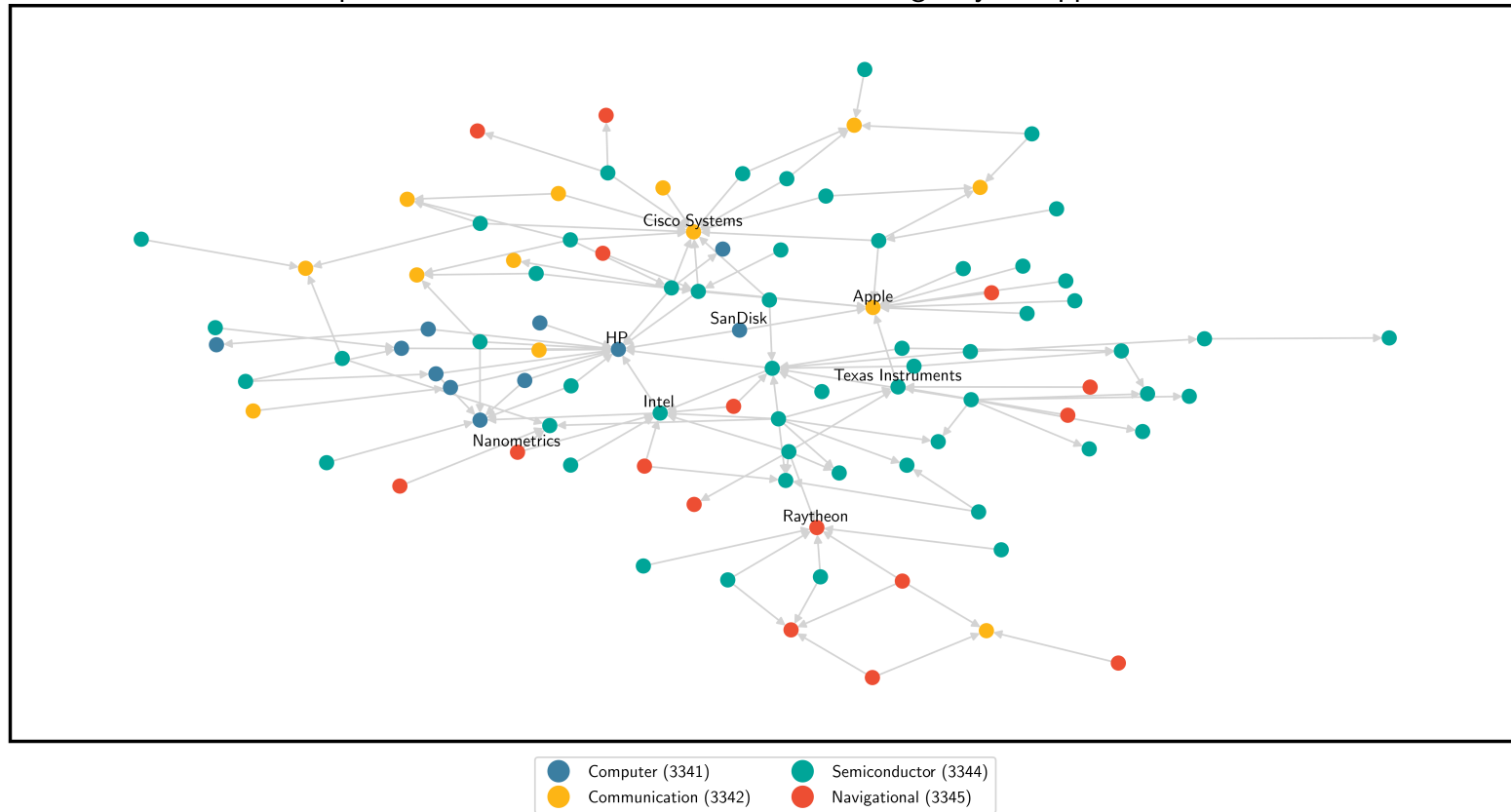
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US Buyer-Supplier Network, 2015



Computer and Electronic Product Manufacturing Buyer-Supplier Network



Two classes of network formation models

Null model: the utility i generates by linking with j depends upon ego (i) and alter (j) attributes alone (attributes may be observed or unobserved).

- Stochastic Block Model
- β -Model

Alternative model: the utility generated by an i to j link *additionally varies* with the presence or absence of *other links* in the network.

- *Strategic* models

Research question

Can we determine whether the network in hand was generated according to null or alternative model?

Very little prior work in this space.

Why I care and you should (might?) too

With strategic behavior:

1. There may be multiple equilibrium network configurations.
2. The observed configuration may not maximize welfare.
3. Vertex removal (and/or local re-wirings) can trigger a process of link revision that is global in scope.

The effect of policies on the form of a network are very different under the null vs. the alternative.

Utility

Random utility framework *a la* McFadden (1973).

Let $\mathbf{d} \in \mathbb{D}$ be an *undirected* adjacency matrix. The utility agent i gets from some feasible network wiring $\mathbf{d}$ is

$$\nu_i(\mathbf{d}, \mathbf{U}; \gamma_0, \delta_0) = \sum_j d_{ij} \left[A_i + B_j + \gamma_0 s_{ij}(\mathbf{d}) - U_{ij} \right],$$

where:

1. A_i is a “extroversion effect”;
2. B_j is a “popularity effect”;

Utility (continued)

1. $s_{ij}(\mathbf{d}) = s_{ij}(\mathbf{d} - ij) = s_{ij}(\mathbf{d} + ij)$ is a network/strategic effect; can be used to model:

(a) “rich-get-richer”: $s_{ij}(\mathbf{d}) = \sum_k d_{jk}$;

(b) transitivity: $s_{ij}(\mathbf{d}) = \sum_k d_{ik}d_{jk}$;

4. $\{U_{ij}\}_{i \neq j}$ idiosyncratic utility-shifter (i.i.d. logistic)

Pelican and Graham (2026, *Restud*) work with a much more general model.

Utility (continued)

The marginal utility for agent i associated with (possible) edge (i, j) is

$$MU_{ij}(\mathbf{D}) = \begin{cases} \nu_i(\mathbf{D}) - \nu_i(\mathbf{D} - ij) & \text{if } D_{ij} = 1 \\ \nu_i(\mathbf{D} + ij) - \nu_i(\mathbf{D}) & \text{if } D_{ij} = 0 \end{cases} \quad (1)$$

where $\mathbf{D} - ij$ is the adjacency matrix associated with the network obtained after deleting edge (i, j) ...

...and $\mathbf{D} + ij$ the one obtained via link addition.

Equilibrium

Network is undirected.

It is convenient to assume utility is transferable.

Use *pairwise stable with transfers* equilibrium concept from Bloch and Jackson (2006).

(PAIRWISE STABILITY WITH TRANSFERS) The network $G(\mathcal{V}, \mathcal{E})$ is pairwise stable with transfers if

$$(i) \forall (i, j) \in \mathcal{E}(G), MU_{ij}(\mathbf{D}) + MU_{ji}(\mathbf{D}) \geq 0$$

$$(ii) \forall (i, j) \notin \mathcal{E}(G), MU_{ij}(\mathbf{D}) + MU_{ji}(\mathbf{D}) < 0$$

Equilibrium (continued)

The marginal utility agent i gets from a link with j is

$$MU_{ij}(\mathbf{d}, \mathbf{U}; \gamma_0, \delta_0) = A_i + B_j + \gamma_0 s_{ij}(\mathbf{d}) - U_{ij}.$$

Pairwise stability then implies that, conditional on the realizations of $\mathbf{A}$, $\mathbf{B}$, $\mathbf{U}$ and the value of externality parameter γ_0 , the observed network must satisfy, for $i = 1, \dots, N - 1$ and $j = i + 1, \dots, N$,

$$D_{ij} = \mathbf{1} \left(\tilde{A}_i + \tilde{A}_j + \gamma_0 \tilde{s}_{ij}(\mathbf{D}) \geq \tilde{U}_{ij} \right) \quad (2)$$

with $\tilde{A}_i = A_i + B_i$, $\tilde{s}_{ij}(\mathbf{D}) = s_{ij}(\mathbf{D}) + s_{ji}(\mathbf{D})$ and $\tilde{U}_{ij} = U_{ij} + U_{ji}$.

Defines a system of $\binom{N}{2} = \frac{1}{2}N(N - 1)$ *nonlinear* simultaneous equations

Equilibrium: Fixed-Point Representation

Consider, similar to Miyauchi (2016), the mapping $\varphi(\mathbf{D}) : \mathbb{D}_N \rightarrow \mathbb{I}\binom{N}{2}$:

$$\varphi(\mathbf{d}) \equiv \begin{bmatrix} \mathbf{1} \left(\tilde{A}_1 + \tilde{A}_2 + \gamma_0 \tilde{s}_{12}(\mathbf{d}) \geq U_{12} \right) \\ \mathbf{1} \left(\tilde{A}_1 + \tilde{A}_3 + \gamma_0 \tilde{s}_{13}(\mathbf{d}) \geq U_{13} \right) \\ \vdots \\ \mathbf{1} \left(\tilde{A}_{N-1} + \tilde{A}_N + \gamma_0 \tilde{s}_{N-1N}(\mathbf{d}) \geq U_{N-1N} \right) \end{bmatrix}. \quad (3)$$

The observed adjacency matrix corresponds to the fixed point

$$\mathbf{D} = \text{vech}^{-1} [\varphi(\mathbf{D})].$$

There may be other $\mathbf{d} \in \mathbb{D}_N$ such that $\mathbf{d} = \text{vech}^{-1} [\varphi(\mathbf{d})]$.

Existence can be shown using Tarski's fixed point theorem for some forms of $s_{ij}(\mathbf{d})$.

Testing goal: challenges

Goal is to construct a test of the no strategic interaction ($\gamma_0 = 0$) null.

Three key challenges:

1. null is composite – nuisance parameter $\delta = \mathbf{A}$ is high dimensional (worry: size distortion);
2. can't evaluate likelihood under the alternative (worry: how to maximize power?);
3. characterizing/simulating null distribution (worry: feasibility).

Testing goal: solutions

1. Apply exponential family theory (Ferguson, 1967; Lehmann & Romano, 2005).
2. Find *locally* best test:
 - (a) derivative of likelihood w.r.t to γ difficult to compute (incompleteness);
 - (b) exploit insights from the econometrics of games (e.g., Tamer, 2003; Bajari *et al.* 2010a,b).
3. Use methods for (constrained) network simulation (e.g., Sinclair, 1993)

Constructing the Test

Under the null we have, for $i = 1, \dots, N - 1$ and $j = i + 1, \dots, N$,

$$\Pr \left(D_{ij} = 1 \mid \mathbf{A} \right) = \frac{\exp \left(\tilde{A}_i + \tilde{A}_j \right)}{1 + \exp \left(\tilde{A}_i + \tilde{A}_j \right)},$$

which corresponds to the β -model of network formation.

Probability of $\mathbf{D} = \mathbf{d}$ takes the exponential family form

$$P_0 \left(\mathbf{d}; \tilde{\mathbf{A}} \right) = c \left(\tilde{\mathbf{A}} \right) \exp \left(\mathbf{d}'_+ \tilde{\mathbf{A}} \right)$$

with $\mathbf{d}_+ = (d_{1+}, \dots, d_{N+})$ equal to the degree sequence of the network.

Constructing the Test (continued)

Let $\mathbb{D}_{N,\mathbf{d}_+}$ denote the set of all undirected $N \times N$ adjacency matrices with degree counts also equal to $\mathbf{d}_+$.

$|\mathbb{D}_{N,\mathbf{d}_+}|$ denotes the size, or cardinality, of this set.

Constructing the Test (continued)

Under H_0 the conditional likelihood of $\mathbf{D} = \mathbf{d}$ given $\mathbf{D}_+ = \mathbf{d}_+$ is

$$P_0(\mathbf{d} | \mathbf{D}_+ = \mathbf{d}_+) = \frac{1}{|\mathbb{D}_{N, \mathbf{d}_+}|}.$$

Under the null of no externalities *all networks with identical degree sequences are equally probable.*

This insight will form the basis of our test.

Constructing the Test (continued)

Let $T(\mathbf{d})$ be some statistic of the adjacency matrix $\mathbf{D} = \mathbf{d}$, say its transitivity index.

Test critical function equals

$$\phi(\mathbf{d}) = \begin{cases} 1 & T(\mathbf{d}) > c_{\alpha}(\mathbf{d}_{+}) \\ g_{\alpha}(\mathbf{d}_{+}) & T(\mathbf{d}) = c_{\alpha}(\mathbf{d}_{+}) \\ 0 & T(\mathbf{d}) < c_{\alpha}(\mathbf{d}_{+}) \end{cases} .$$

We will reject the null if our statistic exceeds some critical value, $c_{\alpha}(\mathbf{d}_{+})$ and accept it – or fail to reject it – if our statistic falls below this critical value.

Constructing the Test (continued)

The critical value $c_\alpha(\mathbf{d}_+)$ is chosen to set the rejection probability of our test under the null equal to α (i.e., to control size).

In order to find the appropriate value of $c_\alpha(\mathbf{d}_+)$ we need to know the distribution of $T(\mathbf{D})$ under the null.

This distribution is straightforward to characterize if we proceed *conditional* on the degree sequence observed in the network in hand.

Constructing the Test (continued)

Under the null all possible adjacency matrices with degree sequence $\mathbf{d}_+$ are equally probable.

The null distribution of $T(\mathbf{D})$ therefore equals its distribution across all these matrices.

By enumerating all the elements of $\mathbb{D}_{N, \mathbf{d}_+}$ and calculating $T(\mathbf{d})$ for each one, we could directly – and exactly – compute this distribution.

In practice this is not (generally) computationally feasible.

Constructing the Test (continued)

If we could efficiently enumerate the elements of $\mathbb{D}_{N, \mathbf{d}_+}$ we would find $c_\alpha(\mathbf{d}_+)$ by solving

$$1 - \alpha = \frac{\sum_{\mathbf{D} \in \mathbb{D}_{N, \mathbf{d}_+}} \mathbf{1}(T(\mathbf{D}) \leq c_\alpha(\mathbf{d}_+))}{|\mathbb{D}_{N, \mathbf{d}_+}|}$$

Alternatively we might instead calculate the p-value:

$$\Pr(T(\mathbf{D}) \geq T(\mathbf{d}) | \mathbf{D} \in \mathbb{D}_{N, \mathbf{d}_+}) = \frac{\sum_{\mathbf{D} \in \mathbb{D}_{N, \mathbf{d}_+}} \mathbf{1}(T(\mathbf{D}) \geq T(\mathbf{d}))}{|\mathbb{D}_{N, \mathbf{d}_+}|}$$

Choosing $T(\mathbf{d})$

Pelican and Graham (2019) show how to choose $T(\mathbf{d})$ to maximize power against local alternatives.

This is hard because one must work with the likelihood of the network under the alternative (which is incomplete).

In practice – as is common with randomization tests – can also pick a test statistic intuitively.

For example $T(\mathbf{d})$ might be the transitivity index.

Testing: Intuition

If the probability that measured transitivity, in a network randomly drawn from the null distribution, lies above observed transitivity is very low...

...we take that as evidence against the β -model and “reject”.

Testing

- This approach to testing is
 - very precise about its description of the null hypothesis;
 - exact.
- We have motivated this test via a particular alternative (and can optimize power vis-a-vis it), but rejection may occur for many reasons.
- ...at minimum the choice of statistic should be guided by researcher intuitions about what departures from the null model are of particular concern.

Sampling from $\mathbb{D}_{N,d_+}$

- Direct enumeration of all the elements of $\mathbb{D}_{N,d_+}$ is generally not feasible.
- Need a method of sampling from $\mathbb{D}_{N,d_+}$ uniformly and also estimating its size.
- We will implement an approximation of the ideal test.

Sampling from $\mathbb{D}_{N,\mathbf{d}_+}$ (continued)

- Blitzstein and Diaconis (2011) develop a sequential importance sampling algorithm for (effectively) uniformly sampling from $\mathbb{D}_{N,\mathbf{d}_+}$
- Two challenges:
 - how to generate a random draw from $\mathbb{D}_{N,\mathbf{d}_+}$;
 - how to do so uniformly (importance weights).

Graphical Integer Sequences

- To construct $\mathbf{D}$ we begin with a matrix of zeros and sequentially add links to it until its rows and columns sum to the target degree sequence.
- Problem is that unless links are added carefully it is easy to get “stuck” (cf., Snijders, 1991).
- The key is to check whether residual degree sequences are graphical as you add links (avoid dead ends).
- $\mathbf{D}_+ = (2, 2, 1)$ is not graphic

Graphical Integer Sequences (continued)

- Erdos and Gallai (1961) showed $\mathbf{D}_+$ is graphical if and only if $\sum_{i=1}^N D_{i+}$ is even and

$$\sum_{i=1}^k D_{i+} \leq k(k-1) + \sum_{i=k+1}^N \min(k, D_{i+}) \text{ for each } k \in \{1, \dots, N\}.$$

Graphical Integer Sequences (continued)

Necessity:

- even: if i is linked to j , then the link is counted in both D_{i+} and D_{j+} .
- For any set S of k agents, there can be at most $\binom{k}{2} = \frac{1}{2}k(k-1)$ links between them (first term).
- For the $N - k$ agents $i \notin S$, there can be at most $\min(k, D_{i+})$ links from i to agents in S .

Graphical Integer Sequences (continued)

Sufficiency of the condition is (evidently) much harder to show.

Erdos and Gallai Theorem provides a simple test for graphicality of a degree sequence.

The next theorem, due to Havel (1955) and Hakimi (1962), shows that this test may be applied recursively.

A Recursive Test

Theorem: (Havel-Hakimi) Let $D_{i+} > 0$, if $\mathbf{D}_+$ does not have at least D_{i+} positive entries other than i it is not graphical. Assume this condition holds. Let $\mathbf{D}_+$ be a degree sequence of length $N - 1$ obtained by

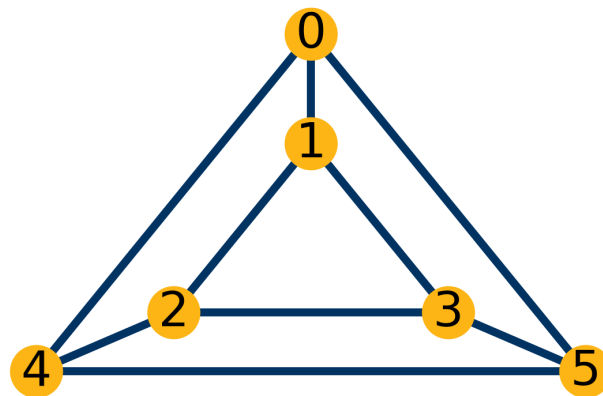
- [i] deleting the i^{th} entry of $\mathbf{D}_+$ and
- [ii] subtracting 1 from each of the D_{i+} highest elements in $\mathbf{D}_+$ (aside from the i^{th} one).

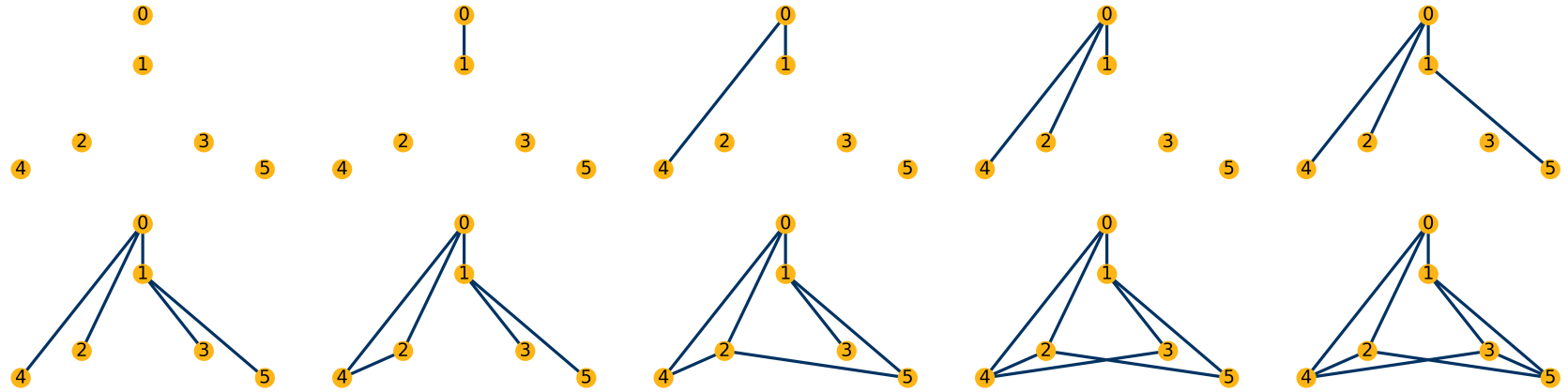
$\mathbf{D}_+$ is graphical if and only if $\mathbf{D}_+$ is graphical. If $\mathbf{D}_+$ is graphical, then it has a realization where agent i is connected to any of the D_{i+} highest degree agents (other than i).

Blitzstein and Diaconis Procedure

- Start with lowest degree agent (with at least one link).
- (Randomly) Link this agent with high degree agents.
- A one is subtracted from the lowest degree agent's, as well as her chosen partners', degree counts.
- Continue until the **residual degree sequence** is zero.

3-regular (i.e., cubic graph)





Blitzstein and Diaconis Procedure (continued)

- Consider the example

$$\begin{aligned}(3, 3, 3, 3, 3, 3) &\rightarrow (2, 2, 3, 3, 3, 3) \rightarrow (1, 2, 3, 3, 2, 3) \rightarrow (0, 2, 2, 3, 2, 3) \\ &\rightarrow (0, 1, 2, 3, 2, 2) \rightarrow (0, 0, 2, 2, 2, 2) \rightarrow (0, 0, 1, 2, 1, 2) \\ &\rightarrow (0, 0, 0, 2, 1, 1) \rightarrow (0, 0, 0, 1, 0, 1) \rightarrow (0, 0, 0, 0, 0, 0) .\end{aligned}$$

- Now imagine that in the 8th step instead of linking agent 3 with agent 4, agents 4 and 5 were linked.

Blitzstein and Diaconis Procedure (continued)

- This would have resulted in a residual degree sequence of $(0, 0, 0, 2, 0, 0)$, which is *not* graphic.
- Algorithm doesn't allow this to occur by checking for whether the residual degree sequence associated with a candidate link is graphical.

Blitzstein and Diaconis Procedure (continued)

- Let $\left(\oplus_{i_1, \dots, i_k} \mathbf{D}_+\right)$ be the vector obtained by adding a one to the $i_1, \dots, i_k$ elements of $\mathbf{D}_+$:

$$\left(\oplus_{i_1, \dots, i_k} \mathbf{D}_+\right)_j = \begin{cases} D_{j+} + 1 & \text{for } j \in \{i_1, \dots, i_k\} \\ D_{j+} & \text{otherwise} \end{cases}$$

- Let $\left(\ominus_{i_1, \dots, i_k} \mathbf{D}_+\right)$ be the vector obtained by subtracting one from the $i_1, \dots, i_k$ elements of $\mathbf{D}_+$:

$$\left(\ominus_{i_1, \dots, i_k} \mathbf{D}_+\right)_j = \begin{cases} D_{j+} - 1 & \text{for } j \in \{i_1, \dots, i_k\} \\ D_{j+} & \text{otherwise} \end{cases}$$

Blitzstein and Diaconis Procedure (continued)

Algorithm: A sequential algorithm for constructing a random graph with degree sequence $\mathbf{d}_+ = (d_{1+}, \dots, d_{N+})'$ is

1. Let $\mathbf{D}$ be an empty adjacency matrix.
2. If $\mathbf{D}_+ = \mathbf{0}$ terminate with output $\mathbf{D}$
3. Choose the agent i with minimal positive degree d_{i+} .
4. Construct a list of candidate partners

$$J = \left\{ j \neq i : \mathbf{D}_{ij} = \mathbf{D}_{ji} = 0 \text{ and } \ominus_{i,j} \mathbf{d}_+ \text{ graphical} \right\}.$$

5. Pick a partner $j \in J$ with probability proportional to its degree in $\mathbf{d}_+$.
6. Set $\mathbf{D}_{ij} = \mathbf{D}_{ji} = 1$ and update $\mathbf{d}_+$ to $\Theta_{i,j}\mathbf{d}_+$.
7. Repeat steps 4 to 6 until the degree of agent i is zero.
8. Return to step 2.

The input for the algorithm is the target degree sequence $\mathbf{d}_+$ and the output is an undirected adjacency matrix $\mathbf{D}$ with $\mathbf{D}'_{\iota} = \mathbf{d}_+$.

Importance Weights

- The Blitzstein and Diaconis (2010) procedure delivers a random draw from $\mathbb{D}_{N, \mathbf{d}_+}$, but not a *uniform* random draw.
- Construct importance weights in order to compute expectations using the correct reference distribution.
- Let $\mathbb{Y}_{N, \mathbf{d}_+}$ denote the set of all possible sequences of links generated by the algorithm given input $\mathbf{d}_+$.

Importance Weights (continued)

- Let $\mathcal{D}(Y)$ be the adjacency matrix induced by link sequence Y .
 - Let Y and Y' be equivalent if $\mathcal{D}(Y) = \mathcal{D}(Y')$.
- We can partition $\mathbb{Y}_{N, \mathbf{d}_+}$ into a set of equivalence classes whose number coincides with the cardinality of $\mathbb{D}_{N, \mathbf{d}_+}$.

Importance Weights (continued)

- Let $c(Y)$ denote the number of possible link sequences produced by the algorithm that produce Y 's end point adjacency matrix.
- Let $i_1, i_2, \dots, i_M$ be the sequence of agents chosen in step 3 of the algorithm in which Y is the output.

Importance Weights (continued)

- Let $a_1, \dots, a_m$ be the degrees of $i_1, \dots, i_M$ at the time when each agent was *first* selected in step 3.
- Blitzstein and Diaconis show that:

$$c(Y) = \prod_{k=1}^M a_k!$$

Importance Weights (continued)

Consider two equivalent link sequences Y and Y' .

Because links are added to vertices by minimal degree (see Step 3), the sequences $i_1, i_2, \dots, i_M$ coincide for Y and Y' .

This means that *the exact same links*, albeit perhaps in a different order, are added at each “stage” of the algorithm (i.e., when the algorithm iterates through steps 4 to 7 repeatedly for a given agent).

The number of different ways to add agent i_k 's links during such a “stage” is simply $a_k!$ and hence $c(Y) = \prod_{k=1}^M a_k!$

Importance Weights (continued)

- Let $\sigma(Y)$ be the probability that the algorithm produces link sequence Y .
- $\sigma(Y)$ is easy to compute:
 - each time a link in step 5 is chosen we record the probability with which it was chosen.
 - this equals the residual degree of the chosen agent divided by the sum of the residual degrees of all agents in the choice set.
 - the product of all these probabilities equals $\sigma(Y)$.

Importance Weights (continued)

We have that $\mathbb{E} \left[\frac{\pi(\mathcal{D}(Y))}{c(Y)\sigma(Y)} \mathbf{1}(T(\mathcal{D}(Y)) > T(\mathbf{d})) \right]$ equals

$$\begin{aligned} &= \sum_{y \in \mathbb{Y}_{N,\mathbf{d}}} \frac{\pi(\mathcal{D}(y))}{c(y)\sigma(y)} \mathbf{1}(T(\mathcal{D}(Y)) > T(\mathbf{d})) \sigma(y) \\ &= \sum_{y \in \mathbb{Y}_{N,\mathbf{d}}} \frac{\pi(\mathcal{D}(y))}{c(y)} \mathbf{1}(T(\mathcal{D}(Y)) > T(\mathbf{d})) \\ &= \sum_{\mathbf{D} \in \mathbb{D}_{N,\mathbf{d}_+}} \sum_{\{y: \mathcal{D}(y)=\mathbf{D}\}} \frac{\pi(\mathbf{D})}{c(y)} \mathbf{1}(T(\mathbf{D}) > T(\mathbf{d})) \\ &= \sum_{\mathbf{D} \in \mathbb{D}_{N,\mathbf{d}_+}} \pi(\mathbf{D}) \mathbf{1}(T(\mathbf{D}) > T(\mathbf{d})) \\ &= \mathbb{E}_\pi [\mathbf{1}(T(\mathbf{D}) > T(\mathbf{d}))]. \end{aligned}$$

Importance Weights (continued)

Here $\pi(\mathbf{D})$ is the probability attached to the adjacency matrix $\mathbf{D} \in \mathbb{D}_{N, \mathbf{d}_+}$ in the target distribution over $\mathbb{D}_{N, \mathbf{d}_+}$.

The ratio $\pi(\mathcal{D}(Y)) / c(Y) \sigma(Y)$ is called the likelihood ratio or the *importance weight*.

We would like $\pi(\mathbf{D}) = 1 / |\mathbb{D}_{N, \mathbf{d}_+}|$ for all $\mathbf{D} \in \mathbb{D}_{N, \mathbf{d}_+}$.

If we set $\pi(\mathbf{D}) = 1$ we see that $\mathbb{E} \left[\frac{1}{c(Y) \sigma(Y)} \right] = |\mathbb{D}_{N, \mathbf{d}_+}|$. This suggests the analog estimator for $|\mathbb{D}_{N, \mathbf{d}_+}|$ of

$$|\hat{\mathbb{D}}_{N, \mathbf{d}_+}| = \left[\frac{1}{B} \sum_{b=1}^B \frac{1}{c(Y_b) \sigma(Y_b)} \right] \quad (4)$$

Importance Weights (continued)

These results suggest we estimate a p-value for our test by

$$\hat{\rho}_{T(\mathbf{G})} = \left[\frac{1}{B} \sum_{b=1}^B \frac{1}{c(Y_b) \sigma(Y_b)} \right]^{-1} \times \left[\frac{1}{B} \sum_{b=1}^B \frac{1}{c(Y_b) \sigma(Y_b)} \mathbf{1}(T(\mathbf{D}_b) > T(\mathbf{d})) \right]$$

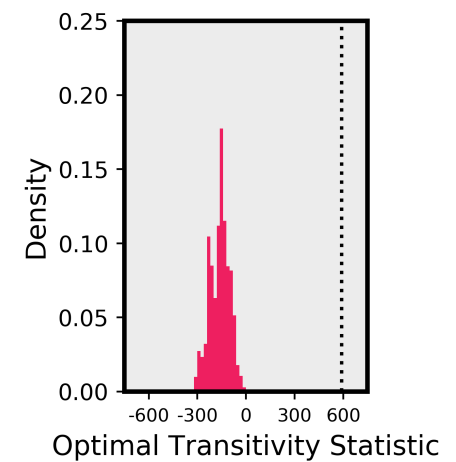
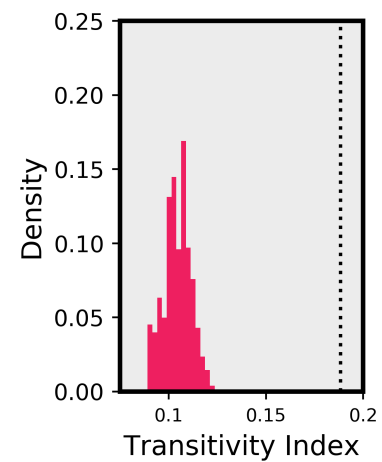
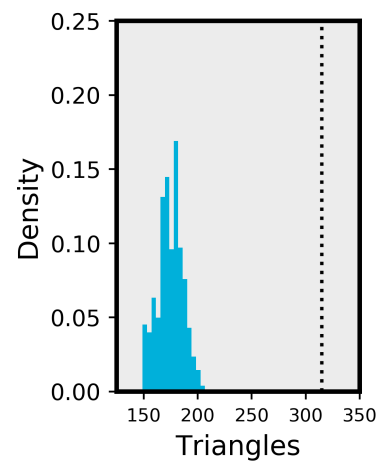
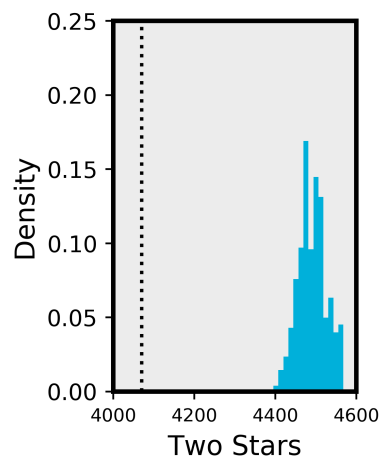
An attractive feature is that the importance weights need only be estimated up to a constant.

This feature is useful when dealing with numerical overflow issues that can arise when $|\mathbb{D}_{N, \mathbf{d}_+}|$ is too large to estimate.

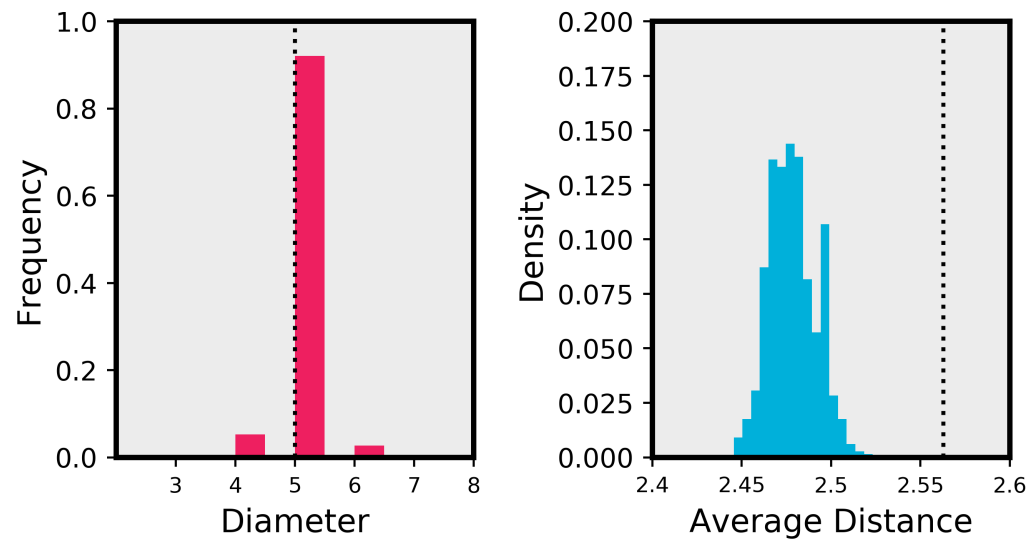
Importance Weights (continued)

- The ratio $\pi(\mathbf{D}(Y)) / c(Y) \sigma(Y)$ is called the likelihood ratio or the **importance weight**.
- Our random network draws are not uniform from the set of interest.
- The importance weights correct for the fact that we are sampling from the wrong distribution.

Nyakatoke Example



Nyakatoke Example (continued)



Wrap-Up

- While using the β -model as a reference model is restrictive it
 - is a natural starting point for hypothesis testing;
 - suggests that an investment in computation skills is likely to be valuable to anyone doing empirical work.
- It might be of interest to condition on additional features of the network in hand...
- ...see Pelican and Graham (2026).